

FARZON NOSIRI

SENIOR SOFTWARE ENGINEER | TECHNICAL LEAD | DISTRIBUTED SYSTEMS | APPLIED AI

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PROFESSIONAL SUMMARY

Accomplished Senior Software Engineer and Technical Lead with 10+ years of experience building scalable distributed systems, backend platforms, and AI-powered applications. Recent work focuses on applied AI engineering, including LLM integration, RAG, agentic workflows, semantic search, OCR model fine-tuning, and speech AI. Proven experience designing high-scale architectures, leading cross-functional engineering teams, and translating business requirements into reliable production systems. Delivered platform transformations supporting B2B growth and performance improvements of approximately 500%, with additional experience in cybersecurity and secure systems engineering.

CORE SKILLS

Languages & Frameworks:	Go, TypeScript/JavaScript, Node.js, Swift/Vapor, Python, Java, React, Angular, Svelte
Architecture:	Distributed systems, multi-tenant systems, microservices, API design, sharding, messaging, scalability, performance optimization
AI & Data:	LLM integration, RAG, agentic workflows, semantic search, embeddings, vector databases, prompt engineering, OCR model fine-tuning, computer vision, STT/TTS, local LLM deployment
Data & Storage:	PostgreSQL, MongoDB, Neo4j, ScyllaDB, MySQL, InfluxDB, S3, MinIO
Infrastructure & Security:	Docker, Linux, CI/CD, NATS, monitoring/observability, network security, VPN/IPsec, incident response
Leadership:	Technical leadership, architecture decisions, cross-functional delivery, stakeholder collaboration, mentorship

PROFESSIONAL EXPERIENCE

Senior Software Engineer | COLORADO SPIRIT HOME CARE, INC.

Jan 2026 - Present

Denver, Colorado

- Design and develop software supporting home-care and adult-day-care operations, including client and caregiver management, care plans, timesheets, EVV workflows, document management, and billing-related processes.
- Build AI-powered applications and integrate LLMs into production software, including RAG-based retrieval, semantic search, metadata enrichment, and agentic workflows.
- Design and develop an AI-assisted digital archive platform with scalable document ingestion, OCR, indexing, and intelligent information retrieval across historical documents and media collections.
- Develop custom training datasets and fine-tune OCR models for handwritten historical documents to improve recognition of handwriting that general-purpose OCR systems struggle to process.
- Integrate and evaluate AI models and services for local LLM inference, document recognition, speech-to-text, and text-to-speech workflows using Ollama, Tesseract, Whisper, and XTTS-v2.
- Develop backend APIs and diagnose issues across application services, databases, AI components, and frontend interfaces.

Technologies: Go, Python, TypeScript, Bun, React, Svelte, PostgreSQL, Ollama, RAG, LLMs, vector databases, Tesseract, OpenCV, Whisper, XTTS-v2, OCR model fine-tuning, STT model fine tuning

Independent Researcher Jan 2024 - Dec 2025

Denver, Colorado

- Authored and published three papers spanning high-load microservice optimization, computational biology, and evaluation of next-generation sequencing software.

Senior Software Engineer / Team Lead | NEXUS TECHNOLOGIES, LLC

Sep 2018 - Dec 2023

Khujand, Tajikistan

- Led the rearchitecture of the company's core communication platform from a B2C product toward a scalable multi-tenant B2B platform using distributed actors, sharding, and message-based services.
- Designed Snowflake-like identifiers incorporating tenant information to support account sharding, workload distribution, and efficient routing across servers.
- Designed an optimized data-synchronization algorithm that improved runtime performance by approximately 500%.
- Developed an MRZ/OCR service for passports and identity documents that automated data extraction, simplified customer onboarding, and contributed to an approximately 40-50% increase in identity-verification applications.
- Built a centralized file service using Node.js and S3-compatible MinIO storage to manage user files across the platform.
- Led and mentored a cross-functional team of approximately 10 backend, frontend, and QA engineers, contributing to architecture decisions, implementation, code quality, and delivery.

Technologies: Swift/Vapor, TypeScript, Node.js, Python, NATS, PostgreSQL, MongoDB, ScyllaDB, Neo4j, InfluxDB, React, Angular, OpenCV, Tesseract, S3/MinIO

Bootcamp Lead | JUNIOR CODE

Feb 2023 - Dec 2023

Khujand, Tajikistan

- Taught Python, algorithms, data structures, software design, and technical-interview skills from beginner through advanced levels.
- More than 40% of students secured internships or job offers with recognized technology companies.

Technologies: Python, TypeScript, Java, Algorithms, Data Structures

Software Developer | STARGROUP

Oct 2017 - Sep 2018

Khujand, Tajikistan

- Designed and implemented backend architecture for DURUD, a CRM platform for small and midsize businesses.
- Built a backend system for a leading cardiology clinic to manage patient appointments, prescriptions, and diagnostic results, improving clinical workflow and doctor-patient interactions.
- Designed the UI/UX for Bank Arvand's mobile-banking application using existing APIs and delivered 122 application screens.
- Developed websites for IMON International and the Osimi Cultural Center and produced source-code documentation for Nexus Technologies, work that contributed to a full-time offer.

Technologies: Java, PHP, MySQL, REST APIs, WordPress, HTML/CSS/JavaScript, Adobe XD, technical documentation**Leading IT Security Specialist | BANK IMON INTERNATIONAL**

Aug 2014 - Oct 2017

Khujand, Tajikistan

- Helped establish and develop the bank's information-security function, strengthening security practices and controls across an organization operating more than 180 branches and offices.
- Designed, installed, and configured a nationwide corporate VPN using IPsec, IKEv2, OpenVPN, Cisco, and MikroTik, providing secure and reliable connectivity for more than 180 branches and offices.
- Contributed to security audits, ISO/IEC 27001-aligned security practices, and disaster-recovery planning to reduce operational and cybersecurity risk.
- Implemented and maintained a secure corporate email platform with backup and redundancy.
- Conducted digital forensics following a security incident involving approximately \$20K in stolen funds and provided evidence that supported law-enforcement efforts to identify the perpetrators.

Technologies: Linux, Kali Linux, OpenVPN, IPsec, IKEv2, Cisco, MikroTik, Postfix, Dovecot, MySQL, Bash, digital forensics**Project Manager & Tester | SENSOTRONICA LTD.**

Nov 2013 - May 2014

Minsk, Belarus

- Managed Agile projects, coordinated technical work with German partners, and tested sensor and Raspberry Pi/GPIO devices used in solar-panel solutions.
- Developed production guidelines for manually manufactured sensor components to improve consistency and quality.

Technologies: Raspberry Pi, GPIO, Node.js, Linux, hardware/software testing**Research Assistant | SCILIFELAB & KTH**

Dec 2012 - Feb 2013

Stockholm, Sweden

- Evaluated BWA, Bowtie, Agile, Stampy, and Mosaik using 20 GB of genome data, including split-read mapping against a reference genome.
- Improved test-data generation performance by rewriting supporting software from Java to C++.

Technologies: Java, C++, BWA, Bowtie, Agile, Stampy, Mosaik, FASTA**Research Assistant | KAROLINSKA INSTITUTET**

May 2011 - Jun 2012

Stockholm, Sweden

- Collaborated on an Epstein-Barr virus research project, analyzing DNA microarray gene-expression data and identifying associated signaling pathways.

Technologies: Java, BioJava, Python, Mosaik, UniProt, FASTA, FunCoup**Software Developer | IT CENTER OF KBTUT**

Nov 2007 - May 2008

Khujand, Tajikistan

- Developed a database-backed application that automated elective-course administration.

Technologies: PHP, JavaScript, MySQL, HTML, CSS**EDUCATION****Master of Science, Computational & Systems Biology | KTH Royal Institute of Technology**

Feb 2013

Stockholm, Sweden

Thesis research investigated signaling pathways involved in Epstein-Barr virus infection and regulation of the OCT-2 transcription factor.

Specialist Diploma, Information Systems in Economics, with Honors

Jul 2009

Technological University of Tajikistan, Khujand Branch

*Khujand, Tajikistan***AWARDS, MEMBERSHIPS & PUBLICATIONS**

- Awards: Software Engineer of the Year, Techno 2023 - NBA "Technologies and Innovation"; Platform of the Year, Innovation Time 2024.
- Membership & Leadership: Senior Member, IEEE; Fellow, Hackathon Raptors; Member, International Association of IT Professionals
- Selected Publications:

"Performance Optimization Techniques for Microservice Architectures in High-Load Scenarios";

<https://doi.org/10.9734/ajrcos/2025/v18i3577>"Regulation of OCT-2"; <http://ijses.com/wp-content/uploads/2025/02/63-IJSSES-V9N1.pdf>

"Evaluation of Software for Next-Generation Sequencing in Mapping."

<https://www.arjonline.org/admin/assets/open-access/evaluation-of-software-of-next-generation-sequencing-in-mapping-165.pdf>